

A Canonical Additive Edge Frame for Zero-Hole Reduced-Residue BDH Remainders

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August 2026

Abstract

We identify the exact additive-frequency geometry of the standard-zero-hole reduced-residue Barban–Davenport–Halberstam remainder. For a prime modulus q , the zero-hole variance is $q^{-1}y^*P_{q-1}y$, where y contains the $q-1$ nonzero additive Fourier coefficients and P_{q-1} projects off their constant direction. The complete-graph Laplacian turns this projection into a tight frame of $(q-1)(q-2)/2$ literal two-frequency differences. The same edge cells distribute the mandatory $(q-2)/(q-1)$ coefficient diagonal exactly, leaving a pure coefficient-off-diagonal remainder in every cell. Four-packet complex polarization is then edgewise, and contraction of the frame recovers the original physical residue kernel with no normalization change. We also prove a scoped uniqueness theorem: every scalar-weighted representation by literal vectors $e_k - e_l$ must use every edge with weight $1/(q-1)$, so no strict edge subset represents the projection. A residue-zero spike refutes separate absolute estimates of equal and unequal frequencies: the two pieces are nonzero, opposite, and sum to zero. An independent exact certificate checks 431 finite identities for $q = 2, 3, 5, 7, 11$. The result is structural: it creates an additive pre-emitter, not a Kloosterman attachment or a prime-shell power saving.

1 Introduction and claim firewall

The prime-dynamics Gate-B reduction studied in the preceding stages produces a signed, prime-modulus, kernel-localized reduced-residue variance. Complex polarization converts its two-sequence scalar into four one-sequence remainders, but the coefficient diagonal must be subtracted with the exact factor $q-2$ before the packet signs are reassembled [5]. Translating a physical block changes the distinguished deleted residue; the moving-hole correction has already been isolated and paid at the critical 1/96 clock [4]. The remaining arithmetic object is therefore the standard-zero-hole remainder itself.

A first additive-Fourier attempt would split that remainder into equal- and unequal-frequency terms and estimate them separately. This order is invalid. For a row supported only at residue zero, the true zero-hole variance is zero, whereas the two proposed pieces have equal nonzero magnitude and opposite sign. Taking absolute values before their cancellation manufactures a large term from an exact zero.

The stable object is the complete-graph Laplacian on the nonzero additive frequencies. It retains the constant-direction cancellation internally and simultaneously distributes the mandatory coefficient diagonal. The paper's contributions are the following.

1. We prove that the zero-hole variance is $q^{-1}y^*P_{q-1}y$ and give its exact complete-graph tight-frame expansion. The frame has rank $q - 2$, $(q - 1)(q - 2)/2$ edge vectors, and redundancy $(q - 1)/2$ for $q > 2$.
2. We prove the edge-mass identity $\sum_e |\Delta_e(n)|^2 = q(q - 2)\mathbf{1}_{q \nmid n}$. It places the $(q - 2)/(q - 1)$ coefficient diagonal inside the same edge cells, so every emitted cell is coefficient-off-diagonal before any estimate.
3. We lift the four-packet polarization edge by edge, recover the frozen physical residue kernel exactly, and expose an oriented (d, k) difference fiber with the unit-annihilating factor $1 - e_q(-dn)$.
4. We prove that a scalar-weighted decomposition by literal two-frequency differences is unique. Every edge is forced, which rules out edge-subset sparsification and identifies the collective source theorem that is still missing.

All four items are finite-dimensional identities proved for every prime q and every finitely supported complex sequence. They do not estimate the edge cells. In particular, the paper does not prove a prime-only Barban–Davenport–Halberstam asymptotic, attach the cells to a Kloosterman theorem, pay the full strict 1/400 Gate-B margin, establish an arithmetic L^2 statement, earn fixed-atom credit, or prove a twin-prime theorem.

2 The frozen zero-hole row

Let q be prime and put

$$e_q(t) = \exp(2\pi it/q), \quad e(t) = \exp(2\pi it).$$

Let $a = (a_n)$ be a finitely supported complex sequence, let $H > 0$, and let $v \in \mathbb{R}$. Write

$$x_n(v) = a_n e(vn/H), \quad A_r(a; v) = \sum_{n \equiv r \pmod{q}} x_n(v). \quad (1)$$

The mean over the $q - 1$ retained residues and the standard-zero-hole variance are

$$\bar{A}^\times = \frac{1}{q-1} \sum_{r \neq 0} A_r, \quad V_0(a; v) = \sum_{r \neq 0} |A_r - \bar{A}^\times|^2. \quad (2)$$

The mandatory coefficient diagonal and its remainder are

$$D_0(a) = \frac{q-2}{q-1} \sum_{q \nmid n} |a_n|^2, \quad R_0(a; v) = V_0(a; v) - D_0(a). \quad (3)$$

The physical Gate-B row carries the outer factor qR_0 , not merely R_0 .

We use the additive transform

$$\hat{A}(k) = \sum_{r \bmod q} A_r e_q(-kr) = \sum_n a_n e(vn/H) e_q(-kn). \quad (4)$$

This convention is fixed throughout. No multiplicative character, inverse residue, or asymptotic estimate enters the frame construction.

3 Zero-hole projection in additive frequency

Set $d = q - 1$, enumerate the nonzero frequencies, and define

$$y = (\hat{A}(k))_{k \in \mathbb{F}_q^\times} \in \mathbb{C}^d, \quad P_d = I_d - \frac{1}{d} \mathbf{1} \mathbf{1}^*. \quad (5)$$

Theorem 3.1 (zero-hole frequency projection). *For every prime q , every finitely supported complex sequence a , and every real v ,*

$$V_0(a; v) = \frac{1}{q} y^* P_d y. \quad (6)$$

Moreover, P_d is an orthogonal projection of rank $q - 2$. For $q = 2$ both sides of (6) vanish.

Proof. For a general row $z = (z_r)_{r \bmod q}$, let $\mu = q^{-1} \sum_r z_r$ and let $V_{\text{all}} = \sum_r |z_r - \mu|^2$. Deleting coordinate zero and recentering the retained coordinates gives the elementary leave-one-out identity

$$\sum_{r \neq 0} \left| z_r - \frac{1}{q-1} \sum_{s \neq 0} z_s \right|^2 = V_{\text{all}} - \frac{q}{q-1} |z_0 - \mu|^2. \quad (7)$$

Indeed, the retained mean equals $\mu - (z_0 - \mu)/(q - 1)$; expanding around μ proves (7).

Apply this identity to $z_r = A_r$. Parseval and Fourier inversion yield

$$V_{\text{all}} = \frac{1}{q} \sum_{k \neq 0} |\hat{A}(k)|^2, \quad A_0 - \mu = \frac{1}{q} \sum_{k \neq 0} \hat{A}(k). \quad (8)$$

Substitution gives

$$V_0 = \frac{1}{q} \sum_{k \neq 0} |\hat{A}(k)|^2 - \frac{1}{q(q-1)} \left| \sum_{k \neq 0} \hat{A}(k) \right|^2, \quad (9)$$

which is (6). The matrix P_d projects onto $\mathbf{1}^\perp$, so its rank is $d - 1 = q - 2$. When $q = 2$, $d = 1$ and $P_d = 0$. $\square$

The null direction in Theorem 3.1 is essential. A vector whose nonzero Fourier coordinates are all equal contributes no zero-hole variance, even if its Euclidean norm is large.

4 The complete-graph tight frame

Let K_d be the complete graph on the nonzero-frequency vertices. For an unordered edge $e = \{k, l\}$ define

$$g_e = e_k - e_l, \quad \Delta_{k,l}(n) = e_q(-kn) - e_q(-ln). \quad (10)$$

The graph Laplacian is

$$\sum_{e \in E(K_d)} g_e g_e^* = dI_d - \mathbf{1} \mathbf{1}^* = dP_d. \quad (11)$$

Theorem 4.1 (canonical additive edge frame). *For every object in Theorem 3.1,*

$$V_0(a; v) = \frac{1}{q(q-1)} \sum_{\{k,l\} \in E(K_{q-1})} \left| \sum_n a_n e(vn/H) \Delta_{k,l}(n) \right|^2. \quad (12)$$

The edge family contains $(q-1)(q-2)/2$ vectors and spans a space of dimension $q-2$. For $q > 2$ its frame redundancy is $(q-1)/2$.

Proof. The edge transform is

$$T_e[a](v) = \langle g_e, y \rangle = \hat{A}(k) - \hat{A}(l) = \sum_n a_n e(vn/H) \Delta_{k,l}(n). \quad (13)$$

Contracting (11) with y gives $\sum_e |T_e[a](v)|^2 = d y^* P_d y$. Theorem 3.1 and $d = q - 1$ prove (12). The count and redundancy follow from the complete graph and the rank of P_d . For $q = 2$ the edge set is empty. $\square$

The redundancy in Theorem 4.1 is exact bookkeeping. It is not a power saving and does not permit a triangle inequality over the edge set. Section 7 will show that every literal edge is forced in a scalar-weighted representation of this projection.

5 Exact edgewise diagonal deletion

Each edge difference annihilates the excluded residue:

$$q \mid n \implies \Delta_{k,l}(n) = 0. \quad (14)$$

The total edge mass on a unit residue is also exact.

Lemma 5.1 (edge mass). *For every integer n ,*

$$\boxed{\sum_{e \in E(K_{q-1})} |\Delta_e(n)|^2 = q(q-2) \mathbf{1}_{q \nmid n}.} \quad (15)$$

Proof. If $q \mid n$, (14) proves the claim. Suppose $q \nmid n$ and put $u(n) = (e_q(-kn))_{k \neq 0}$. Then $\|u(n)\|_2^2 = q - 1$ and $\langle \mathbf{1}, u(n) \rangle = \sum_{k \neq 0} e_q(-kn) = -1$. Therefore

$$\begin{aligned} \sum_e |\langle g_e, u(n) \rangle|^2 &= u(n)^* ((q-1)I - \mathbf{1}\mathbf{1}^*) u(n) \\ &= (q-1)^2 - 1 = q(q-2). \end{aligned}$$

$\square$

Define the edge diagonal and edge remainder by

$$D_e[a] = \sum_n |a_n|^2 |\Delta_e(n)|^2, \quad \mathcal{E}_e^\circ[a](v) = |T_e[a](v)|^2 - D_e[a]. \quad (16)$$

Expanding the square removes the coefficient diagonal inside each cell:

$$\boxed{\mathcal{E}_e^\circ[a](v) = \sum_{t \neq u} a_t \bar{a}_u e(v(t-u)/H) \Delta_e(t) \overline{\Delta_e(u)}.} \quad (17)$$

Corollary 5.2 (zero-hole off-diagonal edge compiler). *The diagonal in (3) distributes over the edge frame without an error:*

$$\frac{1}{q(q-1)} \sum_e D_e[a] = \frac{q-2}{q-1} \sum_{q \nmid n} |a_n|^2 = D_0(a). \quad (18)$$

Consequently,

$$\boxed{R_0(a; v) = \frac{1}{q(q-1)} \sum_e \mathcal{E}_e^\circ[a](v), \quad qR_0(a; v) = \frac{1}{q-1} \sum_e \mathcal{E}_e^\circ[a](v).} \quad (19)$$

Proof. Sum (16) over edges and apply Lemma 5.1. Subtract the result from Theorem 4.1. The second identity retains the mandatory physical outer factor q . $\square$

Corollary 5.2 is the main local improvement over a bare Fourier decomposition. The frame geometry and coefficient-diagonal cancellation now occupy the same cell; neither has to be recovered after an absolute estimate.

6 Polarization and the physical kernel

For two finitely supported sequences β, w , define the bilinear edge cell

$$\mathcal{E}_e^\circ(\beta, w; v) = T_e[\beta](v) \overline{T_e[w](v)} - \sum_n \beta(n) \overline{w(n)} |\Delta_e(n)|^2. \quad (20)$$

It has the pure off-diagonal expansion

$$\mathcal{E}_e^\circ(\beta, w; v) = \sum_{t \neq u} \beta(t) \overline{w(u)} e(v(t - u)/H) \Delta_e(t) \overline{\Delta_e(u)}. \quad (21)$$

Proposition 6.1 (edgewise four-packet polarization). *Let $a^{(j)} = \beta + i^j w$ for $j = 0, 1, 2, 3$. Then, for every edge,*

$$\boxed{\frac{1}{4} \sum_{j=0}^3 i^j \mathcal{E}_e^\circ[a^{(j)}](v) = \mathcal{E}_e^\circ(\beta, w; v).} \quad (22)$$

Proof. For complex x, y , $x\bar{y} = \frac{1}{4} \sum_{j=0}^3 i^j |x + i^j y|^2$. Apply this identity first to $T_e[\beta], T_e[w]$ and then pointwise to $\beta(n), w(n)$. Subtracting the two polarized identities gives (22). $\square$

Let ψ_+ , the prime shell $\mathcal{Q}$, and the literal sequences β, w be the frozen V59 objects. Corollary 5.2, summed before any outer absolute value, gives the exact physical normal form

$$\boxed{\mathfrak{C}_x = \int_{\mathbb{R}} \psi_+(v) \sum_{q \in \mathcal{Q}} \frac{1}{q-1} \sum_{e \in E(K_{q-1})} \mathcal{E}_e^\circ(\beta, w; v) dv.} \quad (23)$$

The ordered block decomposition also commutes with (23): replace β, w by each ordered pair β_b, w_c and sum over (b, c) before applying an absolute value. If the v -integral is evaluated first, its kernel K_H produces the additive pre-emitter

$$\sum_{t \neq u} \beta_b(t) \overline{w_c(u)} K_H(u - t) \Delta_e(t) \overline{\Delta_e(u)}. \quad (24)$$

Equation (24) is not yet a Kloosterman cell.

The following contraction verifies every residue coefficient in (23). For $r, s \bmod q$, put

$$\mathcal{K}_q(r, s) = \sum_e (e_q(-kr) - e_q(-lr))(e_q(ks) - e_q(ls)). \quad (25)$$

Proposition 6.2 (physical-kernel crosswalk). *The edge contraction is*

$$\boxed{\mathcal{K}_q(r, s) = \begin{cases} 0, & rs \equiv 0 \pmod{q}, \\ q(q-2), & r = s \neq 0, \\ -q, & r, s \neq 0, r \neq s. \end{cases}} \quad (26)$$

Hence, on unit residues,

$$\frac{\mathcal{K}_q(r, s)}{q-1} = q \left(\mathbf{1}_{r=s} - \frac{1}{q-1} \right), \quad (27)$$

which is the original V59 coefficient.

Proof. If either residue is zero, every edge difference on that coordinate vanishes. For nonzero r, s , use (11) with $u(r) = (e_q(-kr))_{k \neq 0}$. The diagonal case is $(q-1)^2 - 1 = q(q-2)$ as in Lemma 5.1. If $r \neq s$, additive orthogonality gives $\langle u(s), u(r) \rangle = -1$, while both coordinate sums equal -1 ; contraction with $(q-1)I - \mathbf{1}\mathbf{1}^*$ therefore gives $(q-1)(-1) - 1 = -q$. Dividing by $q-1$ proves (27). $\square$

7 Oriented difference fibers and uniqueness

Write $l = k + d$ in $\mathbb{F}_q$. Counting each unordered edge in both orientations gives

$$\sum_{e \in E(K_{q-1})} \mathcal{E}_e^\circ = \frac{1}{2} \sum_{d \in \mathbb{F}_q^\times} \sum_{\substack{k \in \mathbb{F}_q^\times \\ k \neq -d}} \mathcal{E}_{k, k+d}^\circ. \quad (28)$$

The edge factor separates as

$$\Delta_{k, k+d}(n) = e_q(-kn)(1 - e_q(-dn)). \quad (29)$$

Thus (23) is equivalently

$$\mathfrak{e}_x = \int \psi_+(v) \sum_{q \in \mathcal{Q}} \frac{1}{2(q-1)} \sum_{d \neq 0} \sum_{k \neq 0, -d} \mathcal{E}_{k, k+d}^\circ(\beta, w; v) dv. \quad (30)$$

The base twist k and the unit-annihilating difference factor in (29) are the preferred interface for a future Poisson/Kloosterman compiler. The factor $1/2$ and the exclusion $k = -d$ are mandatory.

The complete edge family cannot be reduced within its literal class.

Theorem 7.1 (literal-edge no-sparsification). *Let $d \geq 1$. Suppose scalar weights $w_{k,l} \in \mathbb{C}$ satisfy*

$$P_d = \sum_{1 \leq k < l \leq d} w_{k,l} (e_k - e_l)(e_k - e_l)^*. \quad (31)$$

Then

$$w_{k,l} = \frac{1}{d} = \frac{1}{q-1} \quad (32)$$

for every edge. In particular, no strict subset of literal two-frequency differences represents P_d with scalar edge weights.

Proof. For $k \neq l$, the (k, l) entry of P_d is $-1/d$. On the right side of (31), only the outer product belonging to $\{k, l\}$ has a nonzero (k, l) entry, and that entry is $-w_{k,l}$. Hence $w_{k,l} = 1/d$ for every pair. With these weights the diagonal entries agree automatically by the complete-graph identity. $\square$

The scope in Theorem 7.1 matters. It excludes neither a dense orthonormal basis of $\mathbf{1}^\perp$, a signed higher-rank cell system, nor an arithmetic theorem that estimates the full redundant frame jointly. It says that deleting literal edges and repairing their weights cannot be the next compiler.

8 Sharp falsifiers and mutation boundaries

8.1 Equal/off-equal estimation is unstable

Take $A_0 = L$ and $A_r = 0$ for every $r \neq 0$. The retained row is zero, so $V_0 = 0$. On the other hand, $\hat{A}(k) = L$ for every $k \neq 0$, and the two terms in (9) are

$$\frac{1}{q} \sum_{k \neq 0} |\hat{A}(k)|^2 = \frac{q-1}{q} |L|^2, \quad (33)$$

$$-\frac{1}{q(q-1)} \left| \sum_{k \neq 0} \hat{A}(k) \right|^2 = -\frac{q-1}{q} |L|^2. \quad (34)$$

Separate absolute estimates lose their exact cancellation. Every edge difference in Theorem 4.1 instead vanishes before an estimate is taken. This fixture refutes the proposed estimate order, not the valid algebraic split in (9).

8.2 One coefficient cancels cellwise

If a has one nonzero coefficient at a unit residue, then $|T_e[a]|^2 = D_e[a]$ for every edge. Thus $\mathcal{E}_e^\circ[a] = 0$ edge by edge. A global diagonal subtraction that leaves nonzero individual cells would fail this local test.

8.3 Mandatory factors

Several nearby formulas describe different objects.

- Summing oriented edges without the factor $1/2$ doubles the scalar.
- Including frequency zero replaces K_{q-1} by K_q and produces an all-residue projection rather than the zero-hole row.
- Replacing $q-1$ by q changes every off-diagonal projection entry and violates Theorem 7.1.
- Omitting $D_e[a]$ returns a positive variance, not the signed coefficient-off-diagonal remainder.
- Dropping one literal edge changes exactly one pair of off-diagonal matrix entries and cannot be repaired by reweighting other edges.

These mutations are included in the exact checker. Their rejection protects the physical signs and normalization; it does not supply arithmetic evidence.

9 Source boundary and related analytic engines

The additive edge frame is an upstream representation theorem. Three nearby analytic sources clarify what remains to be proved.

General-sequence BDH architecture. Harper proves simple Barban–Davenport–Halberstam type asymptotics for broad classes of general sequences under explicit Progressions, Non-concentration, and further structural hypotheses [2]. For prime moduli, grouping reduced residues naturally leads to a zero-hole variance. The source does not state (23), verify its hypotheses uniformly for the four literal prime-dynamics packets, select the prime subset after the signed diagonal subtraction, or perform the ordered block reassembly. The present paper neither invokes nor disproves Harper’s theorem; it gives an exact local normal form for testing a future attachment.

Post-emitter Kloosterman bounds. Blomer and Pascadi obtain strong bounds for bilinear forms with Kloosterman sums, including the critical fixed-modulus saving used in the route ledger [1]. Their theorem accepts an already emitted form with source-valid coefficient arrays. The additive expression (24) has not been transformed into those arrays. In particular, the complete (d, k) frame, four packet signs, prime shell, block norms, and one final reassembly cannot be supplied by renaming (24) a Kloosterman cell.

Exceptional-spectrum and sparse-Fourier machinery. Pascadi’s large-sieve framework for exceptional Maass forms supplies powerful sparse-Fourier and incomplete-Kloosterman tools after the relevant transform and norm hypotheses have been established [3]. Here those hypotheses are precisely part of the open compiler. The source is a plausible downstream engine, not evidence that the edge pre-emitter already meets its input contract.

Bounded novelty claim. The complete-graph Laplacian identity is standard finite-dimensional linear algebra. A bounded arXiv metadata search on 17 August 2026 found no direct match for the combinations “reduced residue and graph Laplacian,” “BDH and additive Fourier,” or “leave-one-out variance and Fourier.” Such a query does not prove literature-wide novelty. The claim made here is narrower: the exact attachment to the frozen zero-hole remainder, edgewise distribution of its mandatory diagonal, physical-kernel crosswalk, and scoped uniqueness of its literal two-frequency representation.

10 Exact computational certificate

The accompanying producer works with Gaussian rationals and prime cyclotomic coefficient vectors. For $q = 2, 3, 5, 7, 11$, it checks the complete-graph Laplacian entrywise, reduces every physical-kernel entry exactly, compares direct and emitted variances on deterministic complex rows, verifies complex polarization, and exercises the spike and edge-weight falsifiers. The independent checker imports neither the producer nor its mathematical module; it reconstructs the projection and closed physical kernel from separate formulas.

The canonical certificate contains 431 exact mathematical rows:

Check family	Exact rows
Laplacian entries, edge counts, and ranks	167
Physical residue kernel	208
Row variance and diagonal normalization	20
Complex polarization	2
Falsifiers and literal-edge uniqueness	20
Explicit mutation checks	14
Total	431

Both programs use explicit failures rather than optimization-sensitive assertions, reject duplicate JSON keys and nonfinite values at their trust boundaries, and reproduce byte-identical output in normal and optimized Python modes. These computations detect transcription and normalization mutations. The proofs in Sections 3–7, not the finite table, establish the general identities.

11 Route consequence and open theorem

Table 1 separates the completed structural layer from the unpaid arithmetic layer.

Table 1: Claim status for the zero-hole additive edge route.

Statement	Status
Zero-hole projection and complete-graph frame	PROVED
Edgewise $(q - 2)$ diagonal deletion	PROVED
Physical-kernel and four-packet crosswalk	PROVED
No strict literal-edge subset representation	PROVED, scoped
Separate absolute equal/off-equal estimates	REFUTED
Whole-frame Poisson/Kloosterman emission	OPEN
Prime-shell fixed saving greater than $1/400$	OPEN / UNPAID
Arithmetic advance, fixed atom, or L^2 theorem	NONE

The smallest next theorem is now explicit.

Open collective compiler. Apply Möbius and Poisson or Voronoi transformations to the complete oriented (d, k) frame in (30) before any edge or fiber triangle inequality. Produce source-valid Kloosterman cells and retain a fixed saving after the ordered blocks, four packet signs, and prime moduli are reassembled once.

The most concrete heuristic is to look for a dual variable shared across the entire tight frame. A shared variable could preserve the complete-graph orthogonality through the transform. If each edge or each d -fiber acquires an independent absolute loss instead, Theorem 7.1 predicts a prohibitive multiplicity and the proposed compiler should be recorded as obstructed rather than promoted conditionally.

12 Conclusion

The standard-zero-hole remainder has a canonical additive representation: the complete-graph Laplacian on nonzero frequencies, with the coefficient diagonal deleted inside each edge cell. This

representation preserves the outer q normalization, the four-packet signs, and the original physical residue kernel. It also explains why the earlier equal/off-equal estimate order fails and why literal edge-subset sparsification cannot repair it.

The resulting object is a precise pre-emitter. The arithmetic burden has not shrunk into a local bound: it has become a joint whole-frame transformation and reassembly theorem. Future progress requires that theorem, or a rigorous obstruction showing that the complete frame cannot share enough dual structure under the available Poisson/Kloosterman engines.

References

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